[External] Bobcat Board Design

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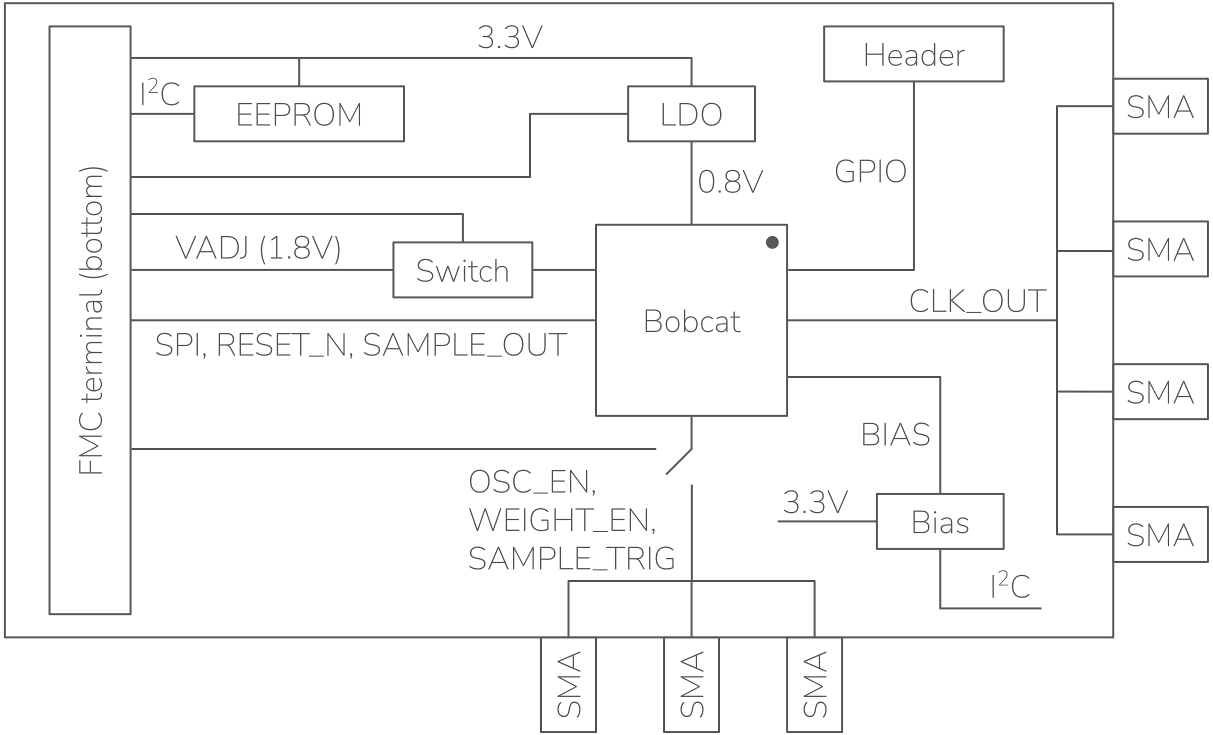
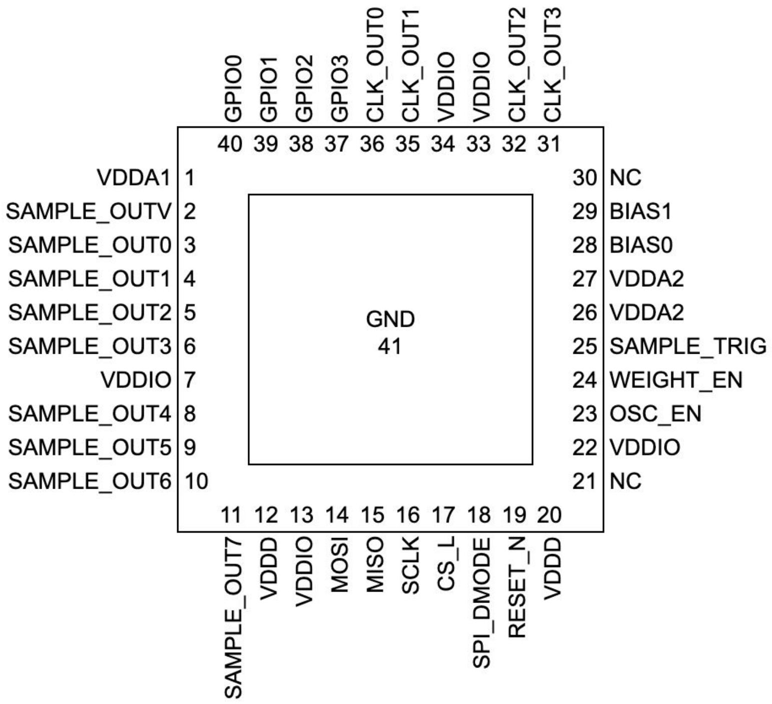
Introduction

- Unconventional AI is rethinking the foundations of a computer to optimize energy efficiency for AI. Founded by experts in AI systems, analog circuits, computing theory, and neuroscience, we are bringing biology-scale efficiency to artificial intelligence.
- We are looking for a partner to collaborate on the board design, fabrication, and assembly of our test chip named Bobcat. The board will plug into the FMC connector of a Genesys 2 platform and extend away from it.

Block Diagram



Bobcat Pinout



- Decoupling capacitors on VDDD, VDDIO, VDDA1, VDDA2
- Series 0Ω on VDDA1, VDDA2
- 10kΩ pull-downs on GPIO0-3, SPI_DMODE, SCLK, MOSI, OSC_EN, WEIGHT_EN, SAMPLE_TRIG
- 10kΩ pull-ups on CS_L, RESET_N
- CLK_OUT0-3 to SMA connectors
- OSC_EN, WEIGHT_EN, SAMPLE_TRIG to SMA connectors with 0Ω resistor options to FPGA
- GPIO0-3 to 1×4 100 mil header

FMC Interface

- Pinout
- VADJ should be connected to Bobcat VDDIO through a load switch and 1×2 jumper
- 3P3V should be connected to EEPROM, LDO, Bias
- LA and HA signals should be connected through series 0Ω resistors
  - SAMPLE_OUTV, SAMPLE_OUT0-7, CS_L, SCLK, MOSI, MISO, SPI_DMODE, RESET_N
  - OSC_EN, WEIGHT_EN, SAMPLE_TRIG with 0Ω resistor options to SMA connectors
  - ANYOUT and EN inputs for LDO
  - Enable signal for load switch
- I^2C SCL/SDA should be connected to the EEPROM and Bias

LDO

- TPS7A8401A
- 3.3V input
- 0.6V to 1.0V output via ANYOUT inputs connected to FPGA
- EN input connected to FPGA with 10kΩ pull-down
- Outputs to Bobcat VDDD, VDDA1, and VDDA2 through 1×2 jumpers

Bias

- Independent programmable current sources for BIAS0 and BIAS1 (nominally 320uA at 0.5V)
- Programmable range of 0-640uA with a step size of ~1uA
- 3.3V supply
- Preferred option
  - I^2C current DAC (default off) connected to the BIASx pin via 1x2 jumper
- Backup option
  - I^2C voltage DAC driving the positive terminal of an op-amp
  - Op-amp output driving the gate of a PMOS
  - PMOS source connected to the negative terminal of the op-amp and 3.3V through a resistor
  - PMOS drain connected to the BIASx pin via 1x2 jumper
  - Needs to be off by default

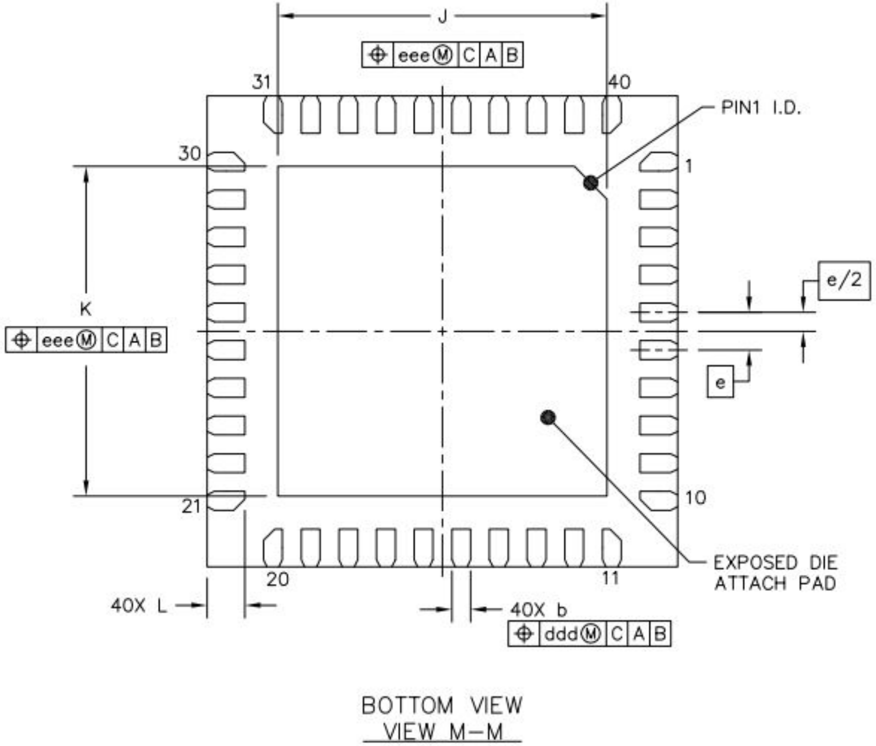
EEPROM

- 3.3V supply voltage
- I^2C
- 8-Kbit

Load Switch

- VADJ (1.2-3.3V)
- Enable input connected to FPGA with 10kΩ pull-down

Bobcat Footprint



		SYMBOL	MIN	NOM	MAX
TOTAL THICKNESS		A	0.8	0.85	0.9
STAND OFF		A1	0	0.035	0.05
MOLD THICKNESS		A2	---	0.65	---
L/F THICKNESS		A3	0.203 REF		
LEAD WIDTH		b	0.15	0.2	0.25
BODY SIZE	X	D	5 BSC		
	Y	E	5 BSC		
LEAD PITCH		e	0.4 BSC		
EP SIZE	X	J	3.4	3.5	3.6
	Y	K	3.4	3.5	3.6
LEAD LENGTH		L	0.35	0.4	0.45
PACKAGE EDGE TOLERANCE		aaa	0.1		
MOLD FLATNESS		bbb	0.1		
COPLANARITY		ccc	0.08		
LEAD OFFSET		ddd	0.1		
EXPOSED PAD OFFSET		eee	0.1		

Note: units are in mm

Socket

- Ironwood Electronics CG25-QFN-2003
- Drawing
    - Holes for screws
    - Keepout areas

Additional Requirements

- Mounting holes at corners furthest from FMC for standoffs
- FMC single width of 69mm
- Target 50Ω impedance for traces going to SMA connectors
- Silkscreen for all reference designators
- Target 4-6 layers with at least 1.6mm thickness
- Test clips for GND

Thank You.

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